

AI-Based Restoration of Degraded Semiconductor Inspection Images

Round 2 Technical Handover Report
Team OOM Survivors - KLA Hackathon 2026

Repository	github.com/GadiMahi/oomsurvivors (branch: v2)
Latest tag	v3-plateau (commit 7d92c6d)
Dataset	4,785 paired SEM images, 256x256 GT from 128x128 NoisyLR
Best validated model	dim48 levels=2, 1.93M params, full-data run, with 4-fold TTA
Headline result	PSNR 23.68 dB / SSIM 0.5151 / LPIPS 0.380 before TTA
vs bicubic baseline	+2.7 dB PSNR, +16.0% SSIM, -28% LPIPS
Inference	9.7 ms/image without TTA, 34.0 ms/image with TTA4
Status	Model work complete; submission packaging outstanding

This report consolidates all Round 2 work: exploratory data analysis, degradation characterisation, every training experiment run, the findings that explain the performance ceiling, and the outstanding items required for submission. Each section is cross-referenced to the git tag or commit where that work landed, so the repository can be checked out at any stage described here.

1. Repository Map and Git History

Work is on branch **v2**. Branch **main** is frozen at the Phase 1 submission state. Three tags mark stable checkpoints in the project's history.

Tag	Commit	What it marks
phase1-submission	510185c	State submitted for Phase 1. Round-1 dataset (natural photographs). Not comparable to Round 2 results
round2-baseline	a58cf68	First complete Round 2 pipeline. Kernel identified as gauss sigma=0.6, noise refit on the full 4,785-pair d
v3-plateau	7d92c6d	Loss experiments complete. Ground-truth noise floor discovered. TTA implemented but not yet measure

1.1 Commit trail, oldest to newest

Commit	Change	Why it mattered
43a9e9f	Fix crop gradient to measure on GT; refit noise constant	Constant selection had been driven by the noise realisation, not image content
45da0fc	GT-only synthetic pool, empirical residual sampling	Unlocked the 3,460 unpaired clean images in the first data drop
4e7b825	Paired-only splits with GT-only training pool	Validation must use real pairs; synthetic cannot validate itself
26aef5	Validate files on load	One corrupt file was crashing every full pass over the dataset
4846450	Clip synthetic targets to [0,1]; merge stats.json	163 of 200 jittered patches were leaving the valid range
190c70e	Non-antialiased kernel candidates; spectral distance	The original low-pass MSE test could not detect kernel differences
4f9f808	Identify downsample kernel as gauss sigma=0.6	Two independent statistics converged on the same value
95b5135	Refit noise against identified kernel	Additive term became non-zero, matching the documented physics
a7857e8	Normalise edge weight map to mean 1	Loss had been silently 2.12x larger than intended
74d1eb6	Require residual bank; fix loss detach	Gaussian fallback was training on an easier problem, silently
4f86ca1	Make model width configurable	Enabled the capacity sweep
a117fe3	run.py --weights flag; width inferred from checkpoint	Needed to evaluate checkpoints of different sizes
0596c27	Add TTA and spectral loss	Measured 72% high-frequency loss in the dim96 model
1e3effa	Configurable U-Net depth; legacy checkpoint removed	Enabled the depth sweep; old checkpoints still load
3c36ba5	Round 2 dim96 baseline checkpoint committed	First model weights preserved in the repository

Key documents in the repository. docs/ROUND2_EDA.md holds the full data analysis. docs/RESULTS.md is the running experiment log. docs/STUDY_GUIDE.md is a 1,570-line technical reference covering the statistics, sampling theory, architecture and losses. artifacts/stats.json carries every measured constant and is the single source of truth for the degradation model.

2. Dataset and Exploratory Analysis

Landed at tag **round2-baseline**. Full detail in docs/ROUND2_EDA.md.

2.1 Composition

Property	Value
Paired images	4,785 (an earlier partial delivery had only 1,325; superseded)
Resolution	GT 256x256, NoisyLR 128x128, every pair exactly 2x
Format	.npy float32, single channel; round-trip verified bit-exact
GT intensity range	[0.0000, 1.0000]
NoisyLR range	[-0.3113, 2.2365] - out-of-range values are intentional
Pixels above 1.0	1.70% mean, 41.1% on the brightest image
Content	SEM micrographs: porous networks, fibres, particulates, films
Source	Derived from NFFA-EUROPE 100% SEM Dataset (CC-BY 4.0)

2.2 Degradation model

The downsampling kernel was recovered from the paired data. An initial low-pass MSE comparison found all candidates within 1.55% and concluded the kernel was unidentifiable. That conclusion was an artefact of the metric: antialiased and non-antialiased kernels differ almost entirely in the high-frequency band, which low-pass filtering discards by construction.

Kernel	Autocorr error	Local-var ratio	Combined score
decimate	-0.0864	1.159	0.6091
gauss sigma=0.4	-0.0594	1.116	0.4448
gauss sigma=0.5	-0.0182	1.045	0.3749
gauss sigma=0.6	+0.0113	1.004	0.2400 (best)
gauss sigma=0.7	+0.0278	0.971	0.2799
area	+0.0500	0.947	0.3334

Two independent statistics - pixel autocorrelation and local texture energy - both cross their target between sigma 0.5 and 0.6. The identified degradation is partial antialiasing: Gaussian blur at sigma about 0.6, then decimation.

2.3 Noise model

```
var(noise | intensity mu) = 2.3807e-02 * mu^2 + 1.0394e-02 * mu + 3.0539e-03
```

The multiplicative term dominates, consistent with speckle. The additive term is non-zero, matching the specification's statement that additive Gaussian noise is present. Notably, when the same fit was run against the *wrong* kernel this term pinned to exactly zero - a parameter agreeing with documented physics that was not fitted to it is stronger evidence than any metric improvement.

Statistic	Measured	Gaussian expectation
Skewness	+0.813	0
Excess kurtosis	+3.919	0
Pixels beyond 3 sigma	0.962%	0.270%
Pixels beyond 5 sigma	0.114%	0.00006%

The noise is strongly heavy-tailed. A Gaussian generator matched to the same variance produces roughly a sixth of the real extreme outliers. The synthetic pipeline therefore resamples 2.64 million real residuals, normalised by predicted sigma and binned by intensity, rather than assuming a parametric family.

3. Experiment Log

All figures measured on the held-out validation split: cluster 3 of a six-way content clustering, 1,165 images the model never trained on. Chosen deliberately as an out-of-distribution proxy, since the hidden test set contains unfamiliar content.

Reference	PSNR	SSIM	edge	flat	LPIPS
bicubic baseline (val_ood)	20.94	0.4441	0.5269	0.4165	0.5248
bicubic baseline (val_id)	20.31	0.5318	0.6239	0.5011	-

Bicubic scores 0.5318 on val_id against 0.4441 on val_ood: cluster 3 is intrinsically harder content. Compare gains over the matching baseline, never raw scores across splits.

3.1 All training runs

Run	Configuration	PSNR	SSIM	LPIPS	ms/img
1	partial data, 70% synthetic, dim64 L1	19.39*	0.5359*	0.3661*	-
2	full data, dim64 L1	23.94	0.5034	0.3655	12.22
3	full data, dim96 L1	23.93	0.5064	0.3610	-
4	LPIPS weight 0.05 -> 0.2	23.66	0.4867	0.3448	-
5	spectral loss weight 3.0	23.46	0.4899	0.3699	-
6	dim48 L2 (depth)	23.89	0.5054	0.3723	9.71
7	dim64 L2 (depth + width)	23.90	0.5074	0.3673	12.27
8	dim48 L2 + TTA4	23.96	0.5178	0.3448	33.96
9	dim48 L2, FULL data, 50 ep - FINAL	23.68	0.5151	0.3798	9.71

* Run 1 used the earlier partial dataset and a different validation split. Its raw numbers are NOT comparable; against its own baseline it scored 1.08 dB *below* bicubic.

3.1a The final run in detail

Run 9 uses the same architecture as run 6 but the full training pool and 50 epochs rather than 40. It is the recommended model. Comparing like with like:

Metric	Run 6 (3,258 pairs)	Run 9 (full data)	Change
PSNR	23.89	23.68	-0.21 dB
SSIM	0.5054	0.5151	+0.0097
SSIM edge	0.5670	0.5836	+0.0166
SSIM flat	0.4849	0.4923	+0.0074
LPIPS	0.3723	0.3798	+0.0075 (worse)

The +0.0097 SSIM gain is the largest single improvement of the entire Round 2 programme - larger than every architecture change combined, and comparable to the TTA gain. It confirms for a third time that real training data volume is the only lever that consistently matters on this problem. The concurrent PSNR loss is a consequence of selecting checkpoints on SSIM: epoch 39 offered PSNR 23.70 with SSIM 0.5133, so the selection rule traded roughly 0.02 dB for 0.002 SSIM. Confirm which metric KLA weights more heavily before treating this as settled.

Training converged genuinely rather than being cut short. Epochs 36 through 50 all fall within 0.0008 SSIM of one another, and training loss moved 0.0021 across those fifteen epochs. Extending to 80 epochs, as earlier configs specified, would have gained nothing. One caveat for whoever revisits this: the selected checkpoint is epoch 37, but epoch 47 recorded identical PSNR, SSIM and edge scores with LPIPS 0.3715 rather than 0.3798 - it dominates epoch 37 on every metric. Single-metric checkpoint selection picked a marginally noisy peak. The difference is small and the saved weights are epoch 37.

Validation caveat. The final run trained on 4,765 of 4,785 pairs, with 20 images held aside for the qualitative comparison figure. Confirm from the run's split file whether the cluster-3 validation set remained excluded from

training. If it did not, run 9's OOD figures are not a clean generalisation estimate and should not be compared directly against runs 2 to 8. The observed direction - PSNR falling rather than rising - argues that the split did hold, since a model evaluated on its own training data would score higher, not lower.

3.2 Test-time augmentation

Transforms	PSNR	SSIM	ms/image
1 (none)	23.773	0.5070	9.68
4 (rotations)	23.955	0.5178	33.96
8 (full dihedral)	23.961	0.5181	66.69

Averaging four rotated predictions gained +0.18 dB and +0.011 SSIM. For scale, every architecture change tried moved SSIM by at most 0.004. Going from 4 to 8 transforms adds +0.0003 SSIM for double the cost and is not worth taking.

3.3 Ensembling

Combination	no TTA PSNR	no TTA SSIM	TTA4 PSNR	TTA4 SSIM
d48L2 alone	23.665	0.5056	23.842	0.5160
+ d64L2	23.809	0.5105	23.908	0.5163
+ lpips02	23.871	0.5113	23.913	0.5137
+ spectral3	23.831	0.5108	23.869	0.5129

Ensembling and TTA are substitutes, not complements. Without TTA a three-model ensemble gains +0.21 dB; with TTA already applied it gains +0.0003 SSIM and can reduce SSIM. Both work by averaging away random error, so once one has done that job the other has nothing left to cancel.

4. Principal Findings

4.1 Synthetic training data hurt, despite passing validation

The first data delivery contained 4,785 clean images but only 1,325 pairs, so a synthetic degradation pipeline was built to use the remainder. It was validated carefully: spatial autocorrelation within 1.2% of real, local texture energy within 0.1%, noise magnitude within 0.4%, out-of-range pixel fraction within 9%.

Training on 70% synthetic data produced a model **1.08 dB below bicubic**. Retraining on real pairs only produced **3.0 dB above** - a 4 dB swing. The effect is confounded with data volume, but PSNR being the worst-affected metric points at degradation mismatch rather than volume alone. The transferable lesson is that matching second-order statistics is a much weaker guarantee than it appears.

4.2 Neither capacity nor receptive field is the bottleneck

Model	Params	Receptive field	PSNR	SSIM
dim64 L1	0.98M	~30 px	23.94	0.5034
dim96 L1	2.14M	~30 px	23.93	0.5064
dim48 L2	1.93M	~60 px	23.89	0.5054
dim64 L2	3.42M	~60 px	23.90	0.5074

Across a 3.5x parameter range and a 2x receptive-field range, every model lands within 0.05 dB and 0.004 SSIM. dim48 L2 and dim96 L1 sit at nearly identical parameter budgets and perform identically, isolating depth from width. Something outside the network sets the ceiling.

4.3 The ground truth carries its own noise floor

The ground truth radial power spectrum flattens above roughly 60% of maximum frequency, with a tail flatness (std/mean) of **0.024**. A flat high-frequency tail is the signature of white noise. Above that point essentially all of the ground truth's high-frequency energy is acquisition noise rather than structure.

This is consistent with how the data was produced: the acknowledgements confirm KLA extracted patches from real SEM captures and added synthetic noise, so the reference images carry their own sensor noise. A fixed fraction of the residual error is therefore irreducible, which explains why capacity, data volume and loss modifications all converged to the same ceiling.

4.4 The model preserves structure; the deficit is band-limited

Reference	PSNR of model output against it
raw ground truth	23.84 dB
ground truth blurred at sigma 0.8	27.87 dB
ground truth blurred at sigma 1.0	28.15 dB
ground truth blurred at sigma 1.2	28.16 dB (peak)
ground truth blurred at sigma 2.0	27.29 dB

The model output matches a low-pass filtered ground truth 4.3 dB better than the raw ground truth, with a clear single peak near sigma 1.1. Structure is therefore preserved correctly - features appear in the right places with the right shapes - and the visible difference is confined to the high-frequency band, which is largely the noise floor described above. Effective resolution is equivalent to a sigma 1.1 low-pass.

4.5 A metric can fail to detect what it was built to measure

Two instances arose. The Round 1 kernel test used low-pass MSE, which discards exactly the band where kernels differ, and concluded the kernel was unidentifiable; a spectrally sensitive test resolved it cleanly. Separately, an edge-versus-overall SSIM heuristic was used to detect over-smoothing; it works on natural photographs where flat regions are featureless, but on SEM images the fine texture lives in exactly those mid-gradient regions, so the heuristic missed a real over-smoothing problem that visual inspection caught immediately.

4.6 Error-reduction methods work; information-extraction methods do not

Sorting every intervention by mechanism produces a clean split. Attempts to extract more information from the input - wider model, deeper model, spectral loss, unsharp masking - all failed, because the information is not present. Attempts to reduce the model's own random error - more real training data, test-time augmentation, ensembling - all succeeded. This is the most useful heuristic to carry forward when deciding what to try next.

5. Approaches Tried and Rejected

Recorded so they are not repeated. Each was a reasonable hypothesis with a measured outcome.

Approach	Rationale	Outcome
Synthetic data at 70%	First delivery had 3,460 unpaired clean images	-0.98 dB below bicubic. Dropped once the full paired set arrived.
Wider model (dim 64 -> 96)	Assumed capacity-limited	+0.003 SSIM, no PSNR change, 37% slower.
Deeper U-Net (1 -> 2 levels)	Receptive field was only ~30 px	+0.004 SSIM. Notably 20% FASTER than the wide model.
LPIPS weight 0.05 -> 0.2	Counter regression-to-mean blurring	-0.27 dB PSNR for -4.5% LPIPS. A trade, not a win.
Spectral (frequency) loss	Model retained only 28% of high-frequency power	Worse on every metric. Raised HF energy without HF structure.
Unsharp masking post-process	Output visibly softer than ground truth	Every setting lost PSNR. Best case +0.0038 SSIM for -0.42 dB.
8-fold TTA	More orientations should average more error	+0.0003 SSIM over 4-fold for double the cost.
Multi-model ensembling with TTA	Independent errors should cancel	Redundant with TTA. Adding models can reduce SSIM.
NFFA-EUROPE external data	21,169 CC-BY SEM images available	Not pursued: usable only via synthetic degradation, which had already been shown

5.1 Bugs found and fixed

Bug	Effect	Commit
Crop gradient measured on noisy LR	Rejection sampling selected on noise, not content	43a9e9f
Edge weight map not normalised	Loss silently 2.12x larger; effective LR rescaled	a7857e8
Dataset RNG created in __init__	All 4 dataloader workers drew identical crops	45da0fc
Synthetic targets unclipped	163/200 jittered patches left the valid [0,1] range	4846450
Gaussian noise fallback silent	Trained on an easier problem without warning	74d1eb6
stats.json overwritten by inventory	Measured constants reverted to placeholders	4846450

6. Current State

6.1 Recommended model

Property	Value
Architecture	NAFNet U-Net, dim=48, levels=2, 1.93M parameters
Training data	4,765 real pairs, 50 epochs, best epoch 37
Loss	edge-weighted Charbonnier + 0.5 Sobel + 0.05 LPIPS(vgg)
Optimiser	AdamW lr 5e-4, cosine to 1e-6, weight decay 1e-4, clip 1.0
Augmentation	D4, CutBlur p=0.5, scale jitter 0.7-1.4x
Inference	4-fold TTA, 33.96 ms/image measured on a T4
Validated PSNR	23.68 dB (bicubic 20.94)
Validated SSIM	0.5151 (bicubic 0.4441)
Validated LPIPS	0.3798 (bicubic 0.5248)
Expected with TTA4	SSIM approx 0.526 - TTA gained +0.011 on the same architecture at run 6 and has not yet been re-measured

6.2 Qualitative assessment

A twenty-image comparison figure (input, bicubic, model, ground truth) was produced on held-out images and is stored with the results. It supports the quantitative picture and should be read alongside it, because the metrics alone understate how much the model achieves. Speckle is removed almost completely across every content type - porous networks, fibres, particulates, smooth films and high-contrast crack structures - with no visible artefacts, ringing or false structure. Bicubic outputs remain visibly noisy in every panel.

The residual difference against ground truth is exactly what section 4.4 predicts: the model output is smoother, and the ground truth carries a fine grain the model does not reproduce. That grain is largely the reference images' own acquisition noise. On the low-contrast panels the difference is most visible; on high-contrast structural content the outputs are close to indistinguishable at viewing scale. Two panels are worth showing to a reviewer specifically: the bright gradient field, where the model recovers a clean smooth surface from an input that is almost entirely speckle, and the crack network, where fine branching structure survives intact.

6.3 Checkpoint inventory

File	Size	Configuration	Location
d48L2_best_nafnet.pt	23 MB	dim48 L2, 3,258 pairs	local Downloads
d64L2_best_nafnet.pt	41 MB	dim64 L2	local Downloads
round2_dim96.pt	26 MB	dim96 L1	repo weights/ (commit 3c36ba5)
lips02_best_nafnet.pt	12 MB	dim64 L1, LPIPS 0.2	local Downloads
spectral3_best_nafnet.pt	12 MB	dim64 L1, spectral 3.0	local Downloads
final_best_nafnet.pt	23 MB	dim48 L2, full data - RECOMMENDED	local Downloads

The dim64 levels=1 full-data checkpoint was lost to a Kaggle session reset; its metrics survive in docs/RESULTS.md. Note that weights/best_nafnet.pt in the repository is a Git LFS pointer to the ROUND 1 model and must be replaced before submission. Training history JSON files for all runs are committed under results/.

6.4 Reproducing the pipeline

```
git clone --branch v2 https://github.com/GadiMahi/oomsurvivors.git
pip install -r requirements.txt

python scripts/run_inventory.py --set data.root=$DATA
python scripts/build_residual_bank.py --set data.root=$DATA --refit --max-images 300
```

```
python scripts/make_cache.py --set data.root=$DATA cache.dir=$CACHE
python scripts/make_splits.py --set data.root=$DATA

python train.py --epochs 50 --set data.root=$DATA cache.dir=$CACHE \
    dataset.synth_p=0.0 model.dim=48 model.levels=2

python run.py <input_dir> <output_dir> --tta 4
```

Order matters: the residual bank must exist before caching, or training falls back to Gaussian noise. Everything is deterministic under seed 1337 - splits and noise fits reproduce exactly. Note that /kaggle/working does not survive session restarts; download checkpoints immediately after each run.

7. Outstanding Work

Model development is complete and further architecture or loss work is not recommended on the evidence above. The remaining items are delivery.

7.1 Submission blockers

- **512x512 forward pass has never been tested.** Every training pair is 256 from 128, but the specification states evaluation may include 512x512 ground truths. Scale jitter (0.7-1.4x) is the only mitigation and is untested at that scale. This is the largest known risk and is a ten-minute check.
- **weights/best_nafnet.pt must be replaced.** It currently holds a Git LFS pointer to the Round 1 model. run.py defaults to this path, so a graded run would silently use the wrong weights. Git LFS is not installed locally.
- **End-to-end verification from a clean clone.** The specification requires the evaluation script to run without manual edits, taking input and output directory arguments.
- **pip freeze environment specification.** Required for reproducibility.
- **Solution presentation.** Twelve-slide structure specified; slides 3 and 5 (dataset analysis, preprocessing and augmentation) can be written directly from sections 2 and 4 of this report.

7.2 Optional, in descending expected value

- **Re-measure TTA4 on the final checkpoint.** TTA is the largest remaining gain available and costs only inference time, but it was measured on run 6's weights, not run 9's. The headline SSIM assumes it transfers. This is a short job and should arguably sit in section 7.1.
- **MS-SSIM loss term.** SSIM is scored by KLA but has never been optimised directly. Expected to trade metrics like the other loss experiments, but untested.
- **Larger training patches (128 rather than 64).** Originally motivated by receptive-field mismatch, which the depth sweep has since disproved. Low expected value.
- **Generative approach (adversarial or diffusion).** Would produce visually sharper texture by synthesising detail rather than recovering it. Expected to cost 1-2 dB PSNR and carries a domain risk: invented texture could read as a defect or mask one. Should not be pursued without guidance on whether hallucinated detail is acceptable in inspection imagery.

7.3 Domain framing worth carrying into the write-up

The physically correct way to remove this noise is frame averaging at acquisition - capture the same field several times and average, so noise falls as the square root of the frame count while structure reinforces. That costs scan time, which is precisely the constraint inspection throughput operates under. Restoration is therefore a software substitute for a physical method the throughput budget will not allow, and its ceiling is set by information theory: the model estimates from a single observation what averaging would have measured directly. The measured ground-truth noise floor is an empirical estimate of where that limit sits.

Bottom line for whoever picks this up. The model is finished and the analysis is documented. Check out tag v3-plateau for the state described in sections 1 to 5, or branch v2 HEAD for the depth sweep and TTA. Read docs/RESULTS.md first - it is the running log and is more current than any summary. The single highest-priority action is testing a 512x512 forward pass, because it is the one failure mode that could invalidate the submission entirely.